SK TRANSWORLD LOGISTICS MALAYSIA

Business Requirements & Solution Specification

Modern B2B Logistics Operations, Customer Portal & Integrated Accounting Platform

Field	Details
Document status	Draft for client validation
Prepared for	SK Transworld Logistics Malaysia
Solution scope	Combined Logistics Management System (CLMS) + Accounting System
Source basis	Client-provided SOFTWARE.pdf (12 pages) and handwritten workflow image
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Purpose
Translate the supplied client workflow and module notes into a structured, implementation-oriented requirements baseline. Source-derived needs are separated from proposed modernisation options and questions that require client confirmation.

CONFIDENTIAL — Working requirements document. Validate operational, tax/e-invoice, accounting and approval details with the client before build sign-off.

Document Control

Item	Description
Owner	SK Transworld Logistics Malaysia
Audience	Client stakeholders, operations, customer service, finance, management, product/design, engineering and QA
Status	Draft — not yet signed off
Change control	Record and approve changes to scope, process, integrations, data fields or approval rules.
Interpretation	Requirements explicitly present in the source are treated as source-derived. Suggested enhancements are labelled proposed. Unspecified policies are listed as open questions.

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1. Executive Summary

SK Transworld Logistics Malaysia intends to modernise logistics and accounting work into a connected B2B web application. The supplied materials describe freight forwarding, customs clearance, haulage/trucking, warehouse operations, storage billing, port/container tracking, job-level document management, a customer portal, billing/accounts, management dashboards and alerts. The handwritten workflow adds quotation and job creation, AI-assisted job-detail entry, supplier costs, payment requests, customer invoicing, finance approval, e-invoice submission, supplier vouchers, bank reconciliation and financial statements.

The central design principle is a single Job Number that connects customer, shipment/container, department activity, digital documents, operational charges, supplier costs, invoices, payment records and job profitability. The system should reduce duplicate data entry, improve shipment visibility, prevent missed billing and make hard-copy records available as searchable digital files.

Recommended solution at a glance

- **One shared job record:** create a job once and reuse its Job Number across departments, documents and financial records.
- **Operations-led charge capture:** capture chargeable forwarding, customs, warehouse, haulage and shipping activities, then create reviewable billing suggestions.
- **Connected finance workflow:** supplier costs, reimbursement/payment requests, customer invoices, e-invoice submission, vouchers and bank reconciliation link back to jobs.
- **Customer self-service:** authorised customers see shipment status, ETA, approved documents and charge visibility.
- **Exception-first management:** highlight overdue payments, pending customs/EDO, free-day expiry, long-stay cargo, missing documents and fleet compliance.
- **AI as assistant:** extract/suggest job details from uploaded documents, with human review before committing important data.

Business outcomes

Outcome	How the solution supports it
Operational visibility	Milestones and statuses by job, shipment, container, customs, warehouse and delivery.
Financial control	Job-level revenue, cost and gross profit; traceable costs/charges; receivables and payables.
Reduced missed billing	Charge suggestions from completed work and alerts for performed but unbilled services.
Document availability	Digital job folder for invoices, BLs, customs files, photos, POD and correspondence.
Customer experience	Portal access to shipment status, ETA, approved documents and charge categories.
Management decisions	Daily operations dashboard and financial summaries by customer, job, shipment and department.

2. Source Understanding and Solution Principles

The client PDF is titled “Combined Logistics System (CLMS) + Accounting System.” It lists 11 main areas and examples of job costing, storage billing, portal visibility, dashboard measures and alerts. The handwritten workflow names customer service, finance and management and sketches the finance-led sequence from quotation through reconciliation and statements.

Source-derived priorities

- Every shipment should have one Job Number linking all departments.
- Job creation should be supported by an AI feature; the exact AI scope is not defined in the source.
- Job costing and profit/loss reporting are important.
- The system should generate and submit e-invoices.
- Quotations and supplier rates/costs should be manageable.
- Financial statements should include profit and loss and balance sheet.
- All hard-copy documents should be managed as digital copies.
- Operations should monitor shipments and follow up to avoid missed billing.

Proposed design principles

- **Single source of truth:** shared job and master records instead of department-specific spreadsheets.
- **Configure rather than hard-code:** make rates, free days, charge rules, status lists, thresholds and approval policies configurable.

- **Human approval for financial commitments:** AI extraction, generated charges and invoices remain reviewable under an agreed approval matrix.
- **Auditability:** preserve history for changes, approvals, invoice events and document versions.
- **Least privilege:** internal users see only permitted data; portal users see only their own organisation's records.
- **Mobile-friendly:** responsive web UI plus driver-oriented trip, status and POD experience.

3. Scope and Boundaries

3.1 In scope

- Freight forwarding: quotation, rate management, booking, BL/HBL/MBL, vessel/voyage, ETA/ETD, FCL/LCL, import/export/transshipment, carrier/co-loader, container tracking, free days, demurrage/detention, milestones and job costing.
- Customs clearance: K1, K2, K3, K4, K5, K8, ZB, DG declaration, DSO, IGM, duty/SST, customs payments, approvals, SSR, inspection, release and document upload.
- Haulage/trucking: truck/trailer/driver masters, container assignment, trip and delivery planning, port appointment, empty pickup, laden delivery, empty return, driver mobile app, GPS tracking, POD, expenses and subcontractors.
- Warehouse: inbound receiving, unloading, cargo count, damage/shortage photos, GRN, palletisation, labelling, stock/location tracking, outbound pick/pack/load/dispatch/POD and storage billing.
- Document repository, customer portal, billing/accounts, supplier costs/payments/vouchers, customer invoices, e-invoice workflow, bank reconciliation, statements, dashboards and alerts.

3.2 Proposed or subject to confirmation

- Integration with a Malaysian e-invoice service/provider; exact route and API are not defined in the source.
- Live carrier, port, customs, GPS, bank and accounting-system integrations. Tracking/reconciliation needs are stated, but providers and interfaces are not.
- AI document extraction and assisted job creation. Supported document types, confidence thresholds and automation boundaries are not specified.
- Email/other notification channels, multi-branch support, multi-currency, tax handling, retention and approval thresholds.

3.3 Not assumed as committed scope

The source does not define payroll, HR, a full CRM/marketing module, e-commerce, a public booking marketplace or a specific third-party accounting product. Do not treat these as included until confirmed.

4. Proposed Users and Access Model

The source explicitly references customer service, finance and management and implies forwarding, customs, haulage and warehouse teams. The role model below is a proposed starting point and requires client validation.

Role	Core responsibilities	Typical permissions (proposed)
System Administrator	Users, access, master data, configuration and integrations	Manage roles/settings/reference data; routine financial approval disabled by default
Management / Director	Oversight, escalation, profitability and performance	Cross-department dashboards/reports; approve exceptional items if policy allows
Customer Service / Job Coordinator	Quotation, job creation, customer coordination and follow-up	Create/edit quotes and jobs; upload documents; update allowed milestones
Forwarding Operator	Bookings, BL/HBL/MBL, carrier details and milestones	Manage forwarding fields and charge suggestions
Customs Operator	Declarations, duty/SST, SSR, inspections and release	Update customs statuses/documents and payment details
Haulage Planner	Fleet, trips, port appointments, dispatch and subcontractors	Plan/assign trips, record costs and monitor POD
Driver / Mobile User	Execute assigned trips and submit evidence	Assigned trip only; status, photo/signature POD and expense submission
Warehouse Operator	Receiving, GRN, inventory movement and dispatch	Manage stock and handling events; capture damage/photos
Finance Executive	Supplier costs, invoices, e-invoice, vouchers and reconciliation	Create draft finance records, reconcile charges and submit for approval

Role	Core responsibilities	Typical permissions (proposed)
Finance Approver	Approve/reject invoices, costs and payment requests	Approve assigned transactions; return with reason; view audit history
Customer Portal User	View own shipments, approved documents and charges	Read-only, customer-scoped access
Auditor / Read-only	Review authorised records and audit trail	Read-only records and reports

Permission requirements

- Enforce access at action and record level; hiding buttons alone is not security.
- Support customer/organisation scoping for portal users and branch/department scoping if required.
- Separate create, edit, submit, approve, post, cancel/void, export and view permissions for financial records.
- Approval users must see preparer, job, evidence and decision history.
- Maintain audit logs for important create/update/status/approval events.
- Where required by policy, prevent users from approving their own transactions.

5. End-to-End Workflow Diagrams

5.1 Operational and finance workflow from the handwritten image

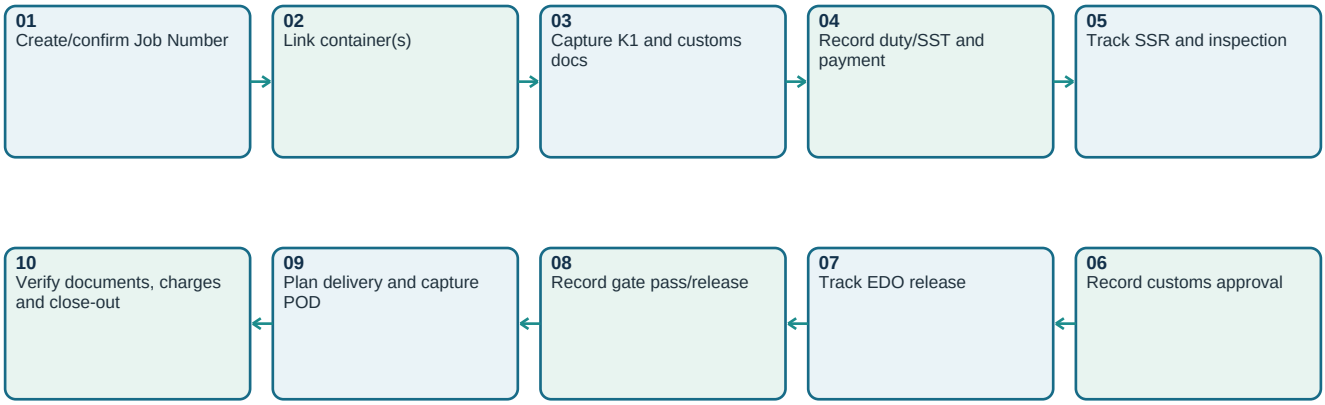
The following diagram preserves the supplied high-level sequence. It does not imply that every accounting posting is automatic; detailed posting rules and exception handling require finance sign-off.



Read left-to-right, then continue on the next row in reverse direction.

Implementation note: the handwritten flow places supplier cost before customer invoice. The proposed solution can support that sequence while allowing a draft invoice from completed operations when a supplier cost is still pending, subject to finance policy.

5.2 Shipment and customs lifecycle



Read left-to-right, then continue on the next row in reverse direction.

The source expresses the sequence as Job → Container → K1 → Duty → SSR → Customs Approval → EDO → Gate Pass → Delivery. K1 is one of several customs document types listed. The additional close-out checkpoint is a proposed control.

5.3 Cross-department hand-offs

Trigger	Expected system behaviour
Quote accepted	Create job from quote or link accepted quote to a new Job Number; carry over customer, service and available rates.
Operational milestone completed	Update job timeline and create eligible charge suggestions from configured rate rules.
Supplier cost received	Attach cost to Job Number/service/container; flag missing or unapproved details.
Ready to bill	Show performed services, charge suggestions, evidence and missing billing items.
Invoice approved	Lock approved invoice version, record approver and submit via configured e-invoice route.
Delivery/POD complete	Make POD searchable and flag missing documents or unbilled work.
Job close-out	Check configurable billing, document and operational/financial exception rules.

6. Functional Requirements

Priority definitions: **Must** = baseline capability from the supplied requirements; **Should** = recommended usability/control improvement; **Could** = optional enhancement. Priorities must be confirmed during workshops.

6.1 Freight Forwarding

Purpose: Manage quotation, booking, shipment details, milestones and forwarding costs.

ID	Requirement	Priority
FF-01	Create customer quotations with service lines, rates, validity and status.	Must
FF-02	Maintain customer and supplier/carrier rates with effective dates and applicable units/currency.	Must
FF-03	Create booking and generate/link a unique Job Number.	Must
FF-04	Capture FCL/LCL, import/export/transshipment, carrier/co-loader, vessel/voyage, ETA/ETD.	Must
FF-05	Maintain BL/HBL/MBL references and attachments.	Must
FF-06	Track containers, free days, demurrage/detention and shipment milestones.	Must
FF-07	Show forwarding revenue, direct costs and job profit/loss.	Must
FF-08	Search by Job No., customer, container, BL, status, vessel and ETA.	Should

Business rules / behaviour

- Job Number is the cross-department key. Preserve milestone history and the rate snapshot applied to each quote/transaction.

6.2 Customs Clearance

Purpose: Track declarations, duty, inspections, approval, release and supporting evidence.

ID	Requirement	Priority
CUS-01	Support K1, K2, K3, K4, K5, K8 and ZB document/type records.	Must
CUS-02	Capture DG declaration, DSO and IGM where applicable.	Must
CUS-03	Track customs status, duty/SST, customs payment and approval.	Must
CUS-04	Track SSR, inspection and release status.	Must
CUS-05	Upload customs documents against job/container.	Must
CUS-06	Show lifecycle from job/container through declaration, duty, SSR, approval, EDO, gate pass and delivery.	Must

Business rules / behaviour

- Customs statuses should be configurable. Uploading a document alone must not be treated as proof that an approval/release occurred.

6.3 Haulage / Trucking

Purpose: Plan trips, manage fleet/subcontractors, capture expenses and calculate profitability.

ID	Requirement	Priority
HAU-01	Maintain truck, trailer and driver master records.	Must
HAU-02	Assign container/job to trips; plan pickup, delivery, port appointment, empty pickup, laden delivery and empty return.	Must
HAU-03	Provide a driver-facing mobile experience for assigned trips and updates.	Must
HAU-04	Support GPS tracking or a GPS provider integration, subject to provider/device availability.	Should
HAU-05	Capture POD with signature/photo and attach it to job/trip.	Must
HAU-06	Record toll, fuel, parking, port, waiting, detention and driver allowance.	Must
HAU-07	Support subcontractor trucking and supplier costs.	Must
HAU-08	Capture haulage rates, customer billing and haulage P/L.	Must

Business rules / behaviour

- The client example is RM1,200 billing – RM700 trucking – RM80 toll – RM50 other cost = RM370 job profit. Actual/estimated costs should be distinguishable.

6.4 Warehouse & Inventory

Purpose: Record inbound receipt, storage, stock movement and outbound dispatch.

ID	Requirement	Priority
WH-01	Link warehouse activity to an existing Job Number or permitted linked job.	Must
WH-02	Record arrival, receiving, unloading, cargo count, damage/shortage, photos and GRN.	Must
WH-03	Capture palletisation, labelling and stock in/out/balance.	Must
WH-04	Store customer, job, container, SKU, carton, pallet, CBM, weight, location/bin/rack, batch/lot and optional expiry.	Must
WH-05	Support delivery order, picking, packing, loading, dispatch and POD.	Must
WH-06	Record handling services: loading/unloading, palletisation, stretch wrap, labelling, forklift, truck loading, documentation, delivery and special handling.	Must
WH-07	Show stock balance and movement history by customer, SKU, location and job.	Should

Business rules / behaviour

- Stock movements should retain source event, operator, timestamp and quantity. Damage/shortage corrections should use an adjustment record, not overwrite the original count.

6.5 Storage Billing & Handling

Purpose: Calculate customer-specific storage charges and chargeable warehouse services.

ID	Requirement	Priority
STO-01	Configure free days and storage charging basis per customer/agreement.	Must
STO-02	Support example: Customer A, 3 free days then RM3/CBM/day.	Must
STO-03	Support example: Customer B, RM8/pallet/day.	Must
STO-04	Support example: Customer C, RM5/carton/month.	Must
STO-05	Generate charge suggestions from storage and handling events.	Must
STO-06	Allow review, reasoned adjustment, approval and transfer to billing.	Should

Business rules / behaviour

- Free-day counting, partial days/months, calendar vs working days, rounding, minimums and exceptions must be configurable and confirmed. Show a calculation breakdown and prevent duplicate billing for the same period/event.

6.6 Port / Container Tracking

Purpose: Provide consolidated container status, ETA and exception visibility.

ID	Requirement	Priority
TRK-01	Search containers and view linked job, customer, milestone, ETA/ETD and last update.	Must
TRK-02	Track free-day expiry and demurrage/detention exposure when data is available.	Must
TRK-03	Allow manual updates and integration-fed events if a source is selected.	Should
TRK-04	Highlight stale tracking data and overdue milestones.	Should

Business rules / behaviour

- Label manually entered, imported and integration-verified events distinctly. Preserve ETA and milestone history.

6.7 Digital Document Management

Purpose: Replace scattered paper files with a searchable digital job folder.

ID	Requirement	Priority
DOC-01	Create one digital folder per Job Number.	Must
DOC-02	Support invoices, packing lists, BL, K1, K2, EDO, LOI, delivery order, gate pass, POD, photos, customs documents and customer emails.	Must
DOC-03	Upload, preview/download, classify and associate files to job/container/trip/invoice.	Must
DOC-04	Flag missing expected documents using configurable checklists.	Should
DOC-05	Control document access and preserve audit/version history for important files.	Must

Business rules / behaviour

- Not every document applies to every job; configure checklists by service/job type. Only explicitly shareable documents should appear in the customer portal.

6.8 Customer Portal

Purpose: Offer secure customer self-service for shipment visibility, documents and charge categories.

ID	Requirement	Priority
PORT-01	Authenticate portal users and scope records to their customer organisation.	Must
PORT-02	Display job/shipment number, container number, current status and ETA.	Must
PORT-03	Provide approved documents such as BL, K1, DO and POD.	Must
PORT-04	Show charges by storage, handling, haulage and forwarding.	Must
PORT-05	Provide shipment list, search and status filters.	Should
PORT-06	Show last-updated time and a contact route for discrepancies.	Should

Business rules / behaviour

- Do not expose internal supplier costs or margins by default. Label charges as estimate/draft/invoiced/paid/overdue according to finance definitions.

6.9 Billing, Accounts & Finance

Purpose: Link operational charges and supplier costs to invoices, vouchers, e-invoice, reconciliation and statements.

ID	Requirement	Priority
FIN-01	Generate reviewable charge lines from forwarding, customs, warehouse, haulage and shipping operations.	Must
FIN-02	Support forwarding fee, documentation, EDI, customs clearance, storage, handling, loading/unloading, pallet, forklift, trucking, toll, waiting, detention, extra stops, freight, BL charges, THC and other carrier charges.	Must
FIN-03	Link each charge, cost, invoice and voucher to Job Number, customer/supplier and department.	Must
FIN-04	Calculate revenue, cost and gross profit by job, customer, shipment and department.	Must
FIN-05	Support supplier cost entry and supplier invoice/payment voucher workflow.	Must
FIN-06	Support reimbursement/payment requests and approval status.	Must
FIN-07	Generate customer invoices from billing lines with finance review/final approval.	Must
FIN-08	Submit e-invoices and retain status/reference/error response from configured service.	Must
FIN-09	Email approved invoice/e-invoice output and log send status.	Must
FIN-10	Support manual vouchers for other bills and agreed supplier-voucher auto-posting.	Must
FIN-11	Support bank reconciliation, P&L; and balance sheet.	Must
FIN-12	Show receivables, overdue customer payments and supplier payables.	Must

Business rules / behaviour

- Keep generated charges as drafts until reviewed/approved. Confirm invoice numbering, credit notes, tax/SST, currencies, chart of accounts, period locks and posting rules. Confirm exact meaning of “auto-post payment voucher.”

6.10 Management Dashboard

Purpose: Summarise daily workload, exceptions and financial performance.

ID	Requirement	Priority
DASH-01	Show deliveries, arriving containers, customs pending, EDO pending, warehouse deliveries and trucks required for today.	Must
DASH-02	Show today’s revenue, month-to-date revenue, outstanding customer payments and supplier payable.	Must
DASH-03	Show job profitability and warehouse, haulage and forwarding revenue under job costing.	Must
DASH-04	Show containers pending/in warehouse, long-stay cargo, customs pending, EDO pending and delivery pending.	Must
DASH-05	Filter/drill down by date, customer, job, department and status.	Should
DASH-06	Export authorised management/finance reports.	Could

Business rules / behaviour

- The source’s counts (e.g. 12 deliveries) are examples, not fixed dashboard values. Define metric formulas, time zone, inclusion criteria and refresh frequency.

6.11 Alerts & Reminders

Purpose: Surface time-sensitive events and missing work before they become operational or financial issues.

ID	Requirement	Priority
ALT-01	Alert before container free time expires.	Must
ALT-02	Alert when EDO is not released.	Must
ALT-03	Alert on overdue customer payment and overdue age.	Must
ALT-04	Alert when warehouse cargo exceeds a configured long-stay threshold (source example: >30 days).	Must
ALT-05	Alert when truck road tax/insurance is due.	Must
ALT-06	Alert on missing documents for applicable jobs/milestones.	Must
ALT-07	Provide alert owner, due date, acknowledgement, status and escalation history.	Should

Business rules / behaviour

- Configure thresholds and channels. Avoid duplicate alerts for one unresolved issue; close an alert with a traceable resolution event.

7. Core Data Model and Business Rules

The following is a proposed logical model derived from the requirement that a Job Number links departments and financial activity. It is not a final physical database schema.

Record / entity	Key relationships and important fields
Customer / Supplier	Organisation/contact details, billing terms, tax details, approved users, supplier/carrier profile, rate agreements.
Quotation / Rate Card	Customer, service type, validity, rate lines, units, currency, exclusions, status and version.
Job	Unique Job Number, customer, service type(s), owner, status, dates, linked quotation, documents, billing state and close-out state.
Shipment / Container	Job Number, container number(s), FCL/LCL, import/export/transshipment, vessel/voyage, BL/HBL/MBL, ETA/ETD and milestones.
Customs Case	Job/container, declaration type/reference, duty/SST, SSR, inspection, approval/release state and evidence.
Trip / Fleet	Job/container, truck, trailer, driver, subcontractor, route/stops, appointments, trip status, expenses and POD.
Warehouse Receipt / Stock Movement	Job/container, SKU, quantity/unit, carton/pallet/CBM/weight, location, batch/lot, damage, GRN and movement history.
Charge / Cost	Job, department, customer/supplier, service/event source, quantity, rate, amount, currency, tax treatment, approval and posting status.
Invoice / Voucher / Payment	Party, Job Number(s), invoice/voucher number, dates, amount, line items, approvals, external e-invoice reference and payment status.
Document / Alert / Audit Event	Job/entity link, type, file/version, access classification; alert owner/due/status; user/time/action/before-after where appropriate.

Cross-cutting business rules

- Every shipment should be linked to one Job Number. A job may have multiple containers, trips, warehouse movements, documents, charges and invoices, subject to confirmed business rules.
- Every charge/cost should identify its source (rate card, operational event, manual entry or integration) and status (draft, reviewed, approved, posted, reversed where relevant).
- Do not silently delete posted finance records; use controlled cancellation, reversal or credit-note processes approved by finance.
- Maintain timestamps consistently and display the business time zone selected by the client.
- Use unique identifiers for jobs, invoices, vouchers and external submission references; numbering formats are client decisions.
- Keep operational status separate from financial status. A delivered job may still be unbilled, unpaid or not fully reconciled.

8. Dashboards, Alerts and Reporting

Dashboard area	Required measures / views	Recommended drill-down
Daily operations	Deliveries; containers arriving; customs pending; EDO pending; warehouse deliveries; trucks required.	Open jobs/containers/trips with owner, date, status and next action.
Financial	Today and MTD revenue; outstanding customer payment; supplier payable; job profitability.	Invoices, receipts/payments, supplier bills and job-cost lines.
Department/job costing	Revenue, cost and gross profit for each job/customer/shipment/department; forwarding, warehouse and haulage revenue.	Source charge/cost, applied rate, approval and posting state.
Warehouse / storage	Stock in/out/balance, long-stay cargo, storage period and charge estimate.	Customer, SKU, location, CBM/pallet/carton, receipt and storage rate.
Alerts	Free time expires, EDO not released, overdue payment, cargo > threshold, vehicle compliance due, missing documents.	Linked record, owner, due date, acknowledgement, escalation and resolution.

Reporting requirements

- Provide date filters and filters by customer, Job Number, department and status where applicable.
- Job profitability must reconcile to the agreed revenue/cost definition; document treatment of tax, pass-through charges and shared costs.
- Export formats, access restrictions, report schedules and retention requirements require confirmation.

• Dashboard cards should be actionable: clicking a metric opens the records contributing to that number.

9. Non-Functional Requirements

Area	Requirement baseline / recommendation
Security	Role-based access, customer data isolation, secure authentication, least privilege and protected secrets. Confirm MFA/SSO needs.
Auditability	Record important user actions, status transitions, approvals, finance posting/reversal and document changes.
Privacy & retention	Define retention, deletion, data residency, privacy notices and handling of customer/driver personal data with the client.
Reliability	Backups, restore tests, error logging and operational monitoring. Availability and recovery targets to be agreed.
Performance	Common job search and detail screens should be responsive under the agreed peak load; set measurable response-time targets during discovery.
Usability	Consistent navigation, clear status labels, searchable lists, inline validation, safe draft handling and responsive layouts.
Accessibility	Keyboard-operable forms, visible focus, readable contrast and clear validation messages.
Integration	Use documented interfaces, retries, idempotency and visible error queues for e-invoice, GPS, carrier, port, customs and bank integrations.
AI governance	Show extracted values and confidence where available; allow human correction; log acceptance/changes; do not auto-approve financial/legal records.
Data integrity	Validate required fields, preserve historical transactions and prevent duplicate charge generation/posting.
Scalability	Design modularly around shared job and finance records; confirm expected users, jobs, files and transaction volumes.
Backup / recovery	Agree RPO/RTO, backup frequency, restore responsibilities and disaster recovery procedure before production.

10. Acceptance Criteria and User Acceptance Testing

The following UAT scenarios form a minimum test baseline. The client should provide realistic sample data and approve expected outcomes before acceptance.

Test ID	Scenario	Pass condition
UAT-01	Create quote and convert to job	Job receives a unique Job Number; customer, service and agreed quote data are linked.
UAT-02	Cross-department traceability	Forwarding, customs, trip, warehouse, documents and finance records can be found through the same Job Number.
UAT-03	AI-assisted job entry	System proposes fields from a supported document; user reviews/corrects before save; changes are traceable.
UAT-04	Customs milestone flow	K1/duty/SSR/approval/EDO/gate pass/delivery statuses show pending steps and history.
UAT-05	Trip and POD	Assigned driver submits status and POD; authorised office user can retrieve evidence from job/trip.
UAT-06	Storage billing	Configured free days and charging basis produce an explainable amount with correct units/dates per approved rules.
UAT-07	Billing completeness	Completed chargeable activities create suggestions; dashboard/alert identifies work not yet billed.
UAT-08	Invoice and e-invoice	Finance reviews/approves invoice; submission status, external reference and errors are retained.
UAT-09	Supplier costs and job P/L	Source costs and revenue roll up to job profitability; test data reconciles to agreed calculation.
UAT-10	Bank reconciliation	Transactions can be matched/unmatched according to agreed workflow; reconciliation state is traceable.
UAT-11	Customer portal isolation	Customer user sees only their own jobs/documents/charges, including when attempting unauthorised direct URLs.
UAT-12	Alerts	Free-day, overdue payment, EDO, long-stay, vehicle compliance and missing-document alerts trigger at configured thresholds.
UAT-13	Audit and permissions	Restricted actions are denied; approvals and important edits show actor/time/history.

Test ID	Scenario	Pass condition
UAT-14	Management reports	Dashboard metrics drill down to source records and financial totals reconcile to agreed reporting rules.

11. Phased Delivery Recommendation

Phase	Scope	Exit criteria
Phase 0 — Discovery & design sign-off	Process workshops, roles/permissions, Job Number rules, accounting/e-invoice path, rate/charge rules, integrations, data migration and report definitions.	Approved BRD, process diagrams, role matrix, integration decisions and acceptance criteria.
Phase 1 — Core job & documents	Customer/supplier masters, quotation/rates, job creation, Job Number, shipment details, shared document folder, milestones and baseline dashboards.	A sample job can be created and tracked across core operations with searchable documents.
Phase 2 — Operations modules	Customs, haulage/trips/POD, warehouse/inventory, storage billing and alerts.	Operational UAT passes; job-linked movements and costs can be traced.
Phase 3 — Finance & customer portal	Charge generation, supplier costs, customer invoices, approval workflow, e-invoice integration, vouchers, bank reconciliation, customer portal and statements.	Finance reconciliation and portal-isolation UAT passes; external integrations verified.
Phase 4 — Optimisation	Advanced tracking integrations, AI document extraction expansion, automation, advanced analytics and additional channels.	Measured improvements are validated; automated actions retain approvals and audit trail.

This sequence is a recommendation, not a commitment to delivery dates. E-invoice and accounting dependencies may affect phasing and must be assessed early.

12. Assumptions, Dependencies and Open Questions

12.1 Assumptions to validate

- A Job Number can be created once per job and linked to one or more containers and multiple service departments; confirm whether actual job granularity differs.
- RM examples are illustrative. Confirm supported currencies and exchange-rate policy.
- Customer-specific storage rates in the PDF are examples and must be verified against real contracts.
- Customer portal users should not see supplier costs or profit margins unless explicitly authorised.
- Email is the minimum implied invoice delivery channel; other notifications are optional until confirmed.
- AI job creation means assisted extraction/suggestion, not unsupervised final approval.

12.2 Decisions required from client

ID	Question / decision	Why it matters
Q-01	What is the exact definition and numbering format of a Job Number? Can one job contain multiple shipments/containers?	Core data model and linking rules.
Q-02	Which departments/services must be active at launch, and which are later phases?	Scope and sequencing.
Q-03	Which accounting software/ledger is currently used? Is this replacing it or integrating with it?	Posting, chart of accounts and reconciliation architecture.
Q-04	Which Malaysian e-invoice provider/API/process must be used? Who owns credentials and compliance decisions?	Integration, validation, submission, rejection and cancellation flows.
Q-05	What are invoice numbering, SST/tax, credit note, multi-currency, exchange-rate and accounting-period rules?	Financial correctness and compliance.
Q-06	What approval matrix applies to customer invoices, supplier bills, payment requests, expenses and adjustments?	Segregation of duties and workflow.
Q-07	Confirm storage billing: free-day counting, start/end dates, calendar vs working days, partial periods, units, minimums and exemptions.	Avoid incorrect automated billing.
Q-08	Which carriers, ports, customs systems, GPS devices, banks and external services need integration? Are APIs available?	Feasibility and costs.
Q-09	What documents are mandatory per job/service type, and which can be shared with customers?	Document checklists and portal security.
Q-10	What exact statuses and hand-offs are used by customer service, forwarding, customs, warehouse, transport and finance?	Workflow configuration.
Q-11	Which AI documents/fields are in scope, and what confidence/review rules are acceptable?	AI accuracy, controls and licensing.
Q-12	How many users, jobs/month, branches, containers, documents and concurrent users are expected?	Performance, storage and deployment sizing.
Q-13	What historical data must be migrated, from which sources, and how much history is required?	Migration plan and data quality.
Q-14	What are required availability, backup, recovery, retention, privacy and security policies?	Production readiness.
Q-15	Which dashboard metrics and financial definitions are authoritative, and who signs them off?	Prevent disagreement over reported figures.

Appendix A. Requirement Traceability

Source material	Requirement areas covered
Handwritten workflow image	Quotation, job creation/cover sheet, AI-assisted job details, payment request/reimbursement, supplier cost, customer invoice, finance approval, e-invoice, email delivery, supplier voucher, other-expense voucher, bank reconciliation, statements, dashboard, shipment follow-up and missed-billing prevention.
SOFTWARE.pdf pp. 1–2	Freight forwarding; Job Number; customs clearance types, duties, SSR, inspection, release and document uploads.
SOFTWARE.pdf pp. 3–4	Truck/trailer/driver, trip planning, driver app, GPS, POD, trip expenses, subcontractor trucking and warehouse inbound.
SOFTWARE.pdf pp. 5–6	Warehouse storage fields, outbound operations, automated storage billing examples and handling charges.
SOFTWARE.pdf pp. 7–8	Port/container tracking, per-job digital document folder and customer portal.
SOFTWARE.pdf pp. 9–10	Operational charge categories, revenue/cost/gross profit, operational and financial dashboard measures.
SOFTWARE.pdf pp. 11–12	Alerts for free time, EDO, overdue payments, long-stay cargo, vehicle documents and missing documents.

Sign-off recommendation: Review this document with customer service, forwarding, customs, haulage, warehouse, finance and management. Resolve open questions and approve the workflow diagrams, role/permission matrix, financial rules, integration scope and UAT scenarios before development begins.

End of document — Version 1.0, Draft for client validation